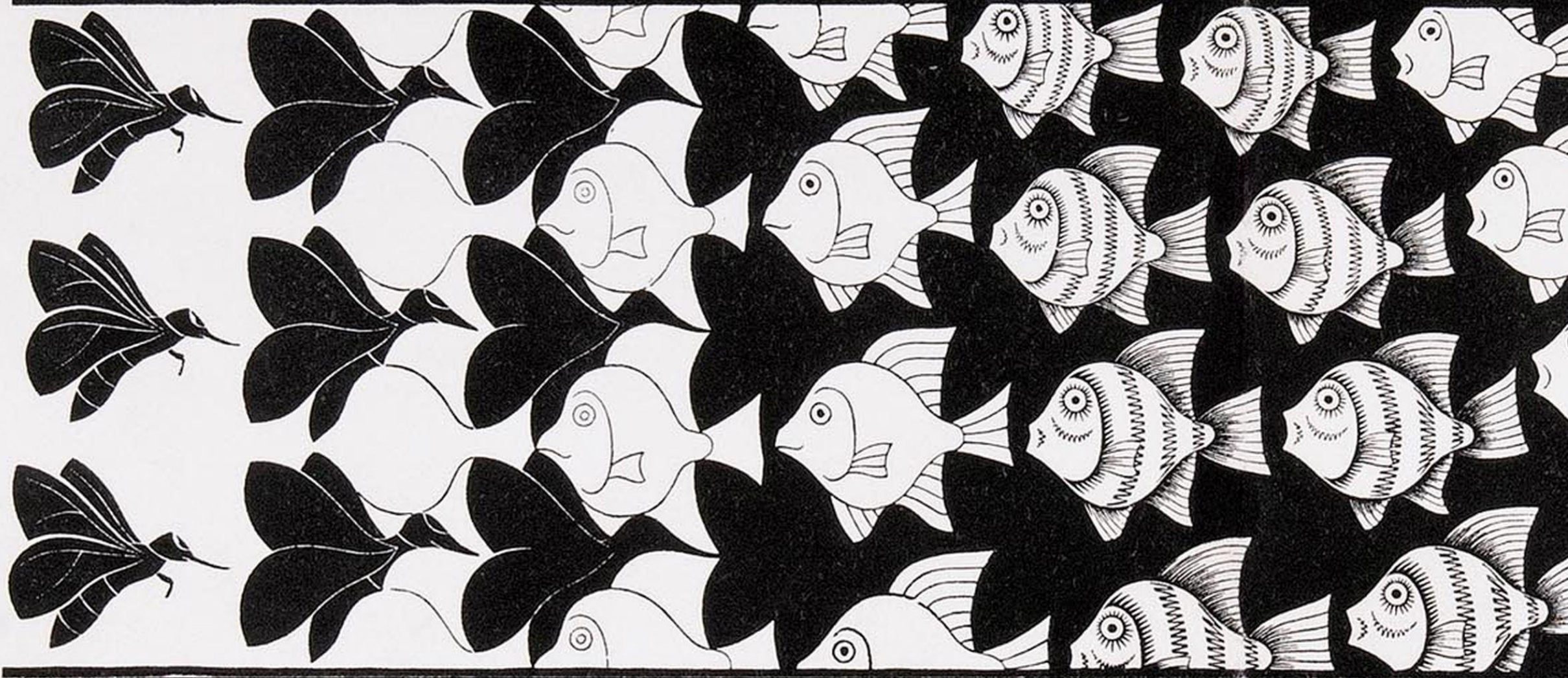


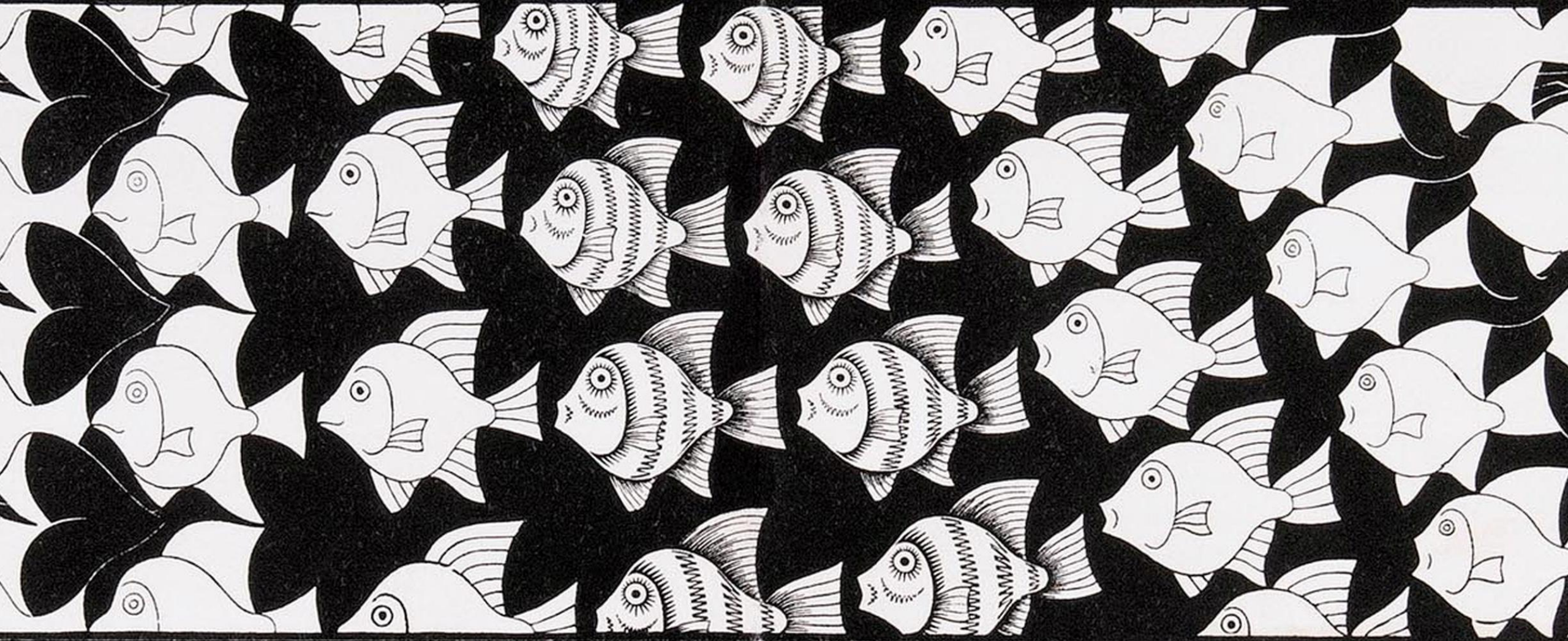


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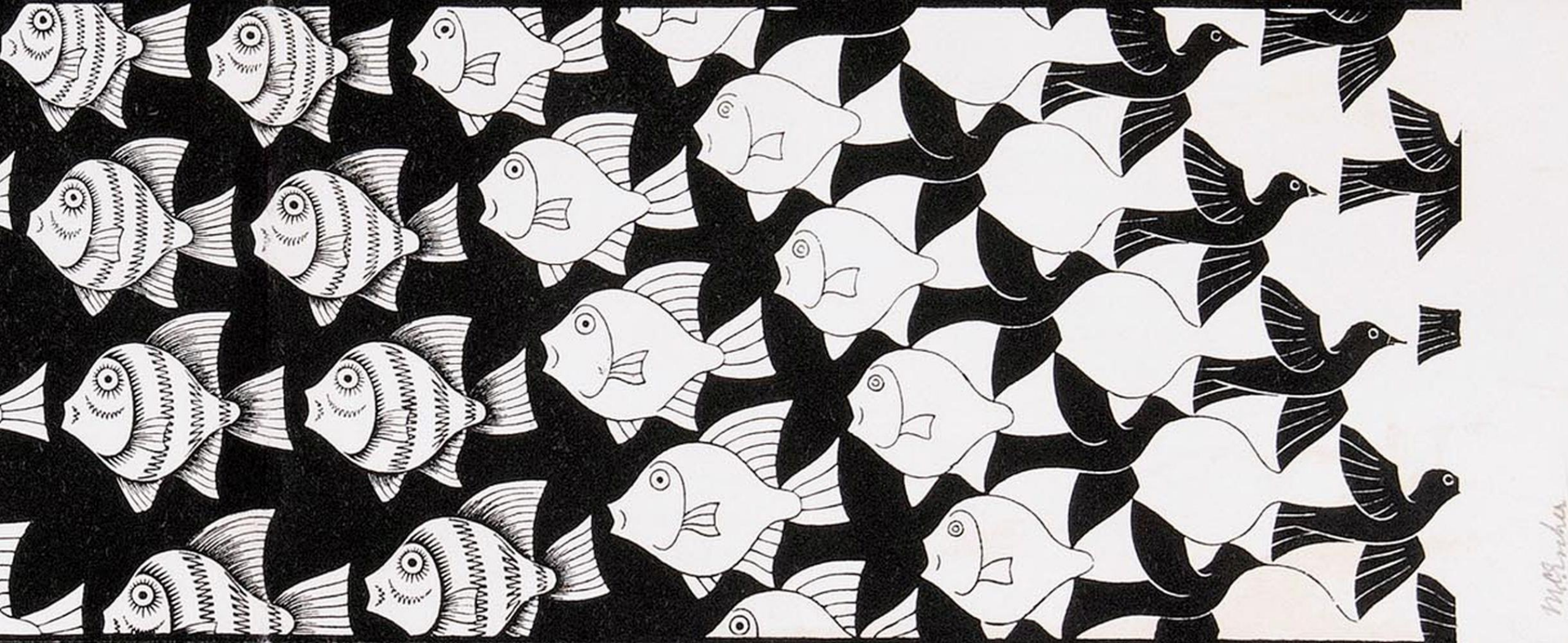
30th Apr 2026

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Gravity2026@IBS



Holomorphic structure of massive
scalar fields in dS_2



based on 2601.20947 in collaboration with Lukas W. Lindwasser and Massimo Porrati

Quantum field theory in de Sitter?

Relevant for cosmology, but **not well understood!**

→ This talk: Toy model in **two dimensions**

Transformations on Euclidean plane $\mathbb{R}^2 \simeq \mathbb{C}$

$$z \mapsto f(z, \bar{z})$$

→ **conformal** = **analytic**

[Belavin, Polyakov, & Zamolodchikov '84]

Posterchild example of CFT_2 : **Massless** scalar field on flat space.

$$S_{\text{flat}} = \int dz d\bar{z} \partial\phi \bar{\partial}\phi$$

Shifts $\phi \mapsto \phi + c$ generated by invariant currents

$$j(z, \bar{z}) \equiv \partial\phi, \quad \bar{j}(z, \bar{z}) \equiv \bar{\partial}\phi$$

On-shell: Conservation = **holomorphicity**

$$\bar{\partial}j = 0, \quad \partial\bar{j} = 0$$

→ Classical phase space **decomposes**

$$\phi(z, \bar{z}) = \varphi(z) + \tilde{\varphi}(\bar{z})$$

→ 2d factorisation “miracle”

In this talk: **Massive** scalar field on (A)dS₂

$$S_{\text{massive}} = -\frac{1}{2} \int d^2x \sqrt{-g} (g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + m^2 \phi^2)$$

Develops **shift symmetries**

$$\delta\phi = S_{A_1 \dots A_k} X^{A_1} \dots X^{A_k}, \quad \text{traceless \& symmetric } S$$

[Bonifacio, Hinterbichler, Joyce, & Rosen '18]

[Anninos, Anous, Pethybridge, & Şengör '23]

for

$$m^2 = -k(k+1)H^2 \quad \longrightarrow \quad \Delta = k+1, \quad k \in \mathbb{Z}_{\geq 0}$$

[Joung, Mourad, & Parentani '07]

[Bros, Epstein, & Moschella '10]

Associated with invariant **currents**

...

$$F(z, \bar{z}) \equiv \partial^{k+1} \phi + \dots, \quad \bar{F}(z, \bar{z}) \equiv \bar{\partial}^{k+1} \phi + \dots$$

Conservation = **holomorphicity**

$$\bar{\partial}F = 0, \quad \partial\bar{F} = 0$$

→ **Suggestive...?**

canonical quantisation

[Birrell & Davies '82]

[Wald '94]

...

Wave equation

First: Construct **classical phase space** of solutions.

Conformally compactified dS_2 w/ conformal time

$$ds^2 = \frac{1}{H^2 \cos(\eta)^2} (-d\eta^2 + d\vartheta^2)$$

→ E.o.m.: Wave equation

$$\left(-\partial_\eta^2 + \partial_\vartheta^2 + \frac{k(k+1)}{\cos(\eta)^2} \right) \phi = 0$$

Admits natural **KG inner product**

$$\left(\phi^{(1)}, \phi^{(2)} \right) = i \int_{-\pi}^{\pi} d\vartheta \left(\overline{\phi^{(1)}} \partial_\eta \phi^{(2)} - \partial_\eta \overline{\phi^{(1)}} \phi^{(2)} \right)$$

Mode solutions

$$\phi(\eta, \vartheta) = \phi_{\text{phys.}}(\eta, \vartheta) + \underbrace{\phi_{\text{pos.}}(\eta, \vartheta) + \phi_{\text{mom.}}(\eta, \vartheta)}_{\phi_0(\eta, \vartheta)}$$

[Higuchi, Schmieding, & Blanco '21]

Physical solutions spanned by

...

$$\{f_n = \psi_{|n|}(\eta)e^{in\vartheta} \mid |n| \geq k+1\}$$

$$(f_m, f_n) = \delta_{m,n}$$

Zero-norm solutions spanned by

$$\{g_n^{(x)}, g_n^{(p)} \mid |n| \leq k\}$$

$$\left(g_m^{(x/p)}, g_n^{(x/p)}\right) = 0, \quad \left(g_m^{(x)}, g_n^{(p)}\right) = i\delta_{m,n}$$

Mode expansions

Physical modes

$$\phi_{\text{phys.}} = \sum_{|n| \geq k+1} a_n f_n + a_n^\dagger \overline{f_n}$$

$$a_n = (f_n, \phi), \quad a_n^\dagger = (\overline{f_n}, \phi) \quad \longrightarrow \quad [a_n, a_n^\dagger] = 1$$

Zero-norm states

$$\phi_{\text{pos.}} = \sum_{|n| \leq k} x_n g_n^{(x)}, \quad \phi_{\text{mom.}} = \sum_{|n| \leq k} p_n g_n^{(p)}$$

$$x_n = -i \left(\phi, g_n^{(p)} \right), \quad p_n = i \left(\phi, g_n^{(x)} \right) \quad \longrightarrow \quad [x_m, p_n] = i \delta_{m+n, 0}$$

Canonical quantisation

Quantisation = choice of complex structure on space of **finite-norm** solutions

$$\mathcal{S} = \mathcal{S}_- \oplus \mathcal{S}_+$$

Some **subtleties** (already present for massless!)

[Allen '85; Allen & Folacci '87]
[Kirsten & Garriga '93]
[De Bievre & Renaud '98]
[Tolley & Turok '01]
...

- Shift symmetries → **zero modes** from

symmetry of action: $\delta\phi = \phi_{\text{pos.}}$

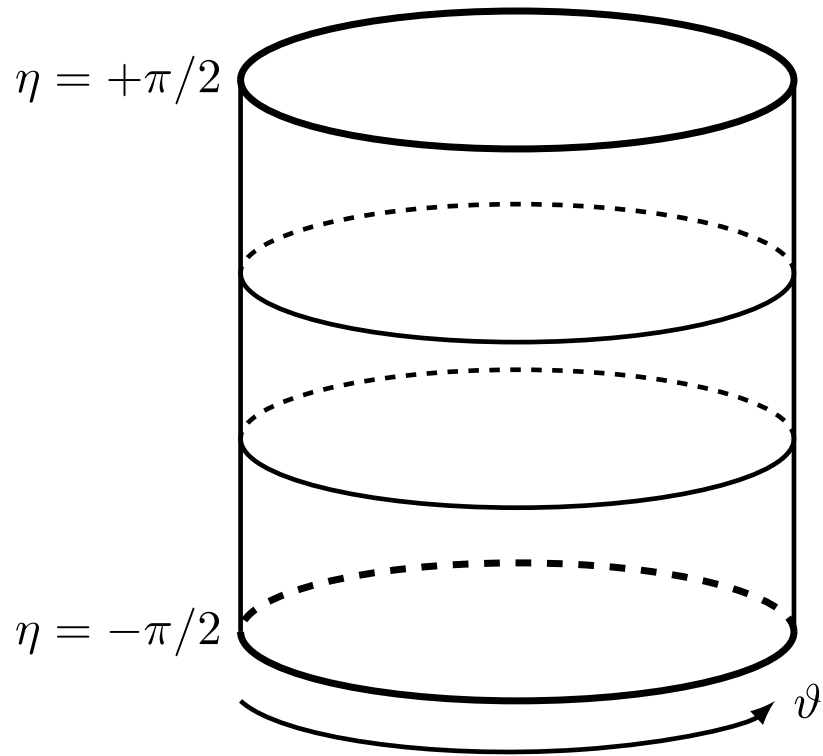
→ Impose equivalence relation on operators $\phi \sim \phi + \phi_{\text{pos.}}$

- de Sitter isometries mix finite- and zero-norm states: Restrict physical states to

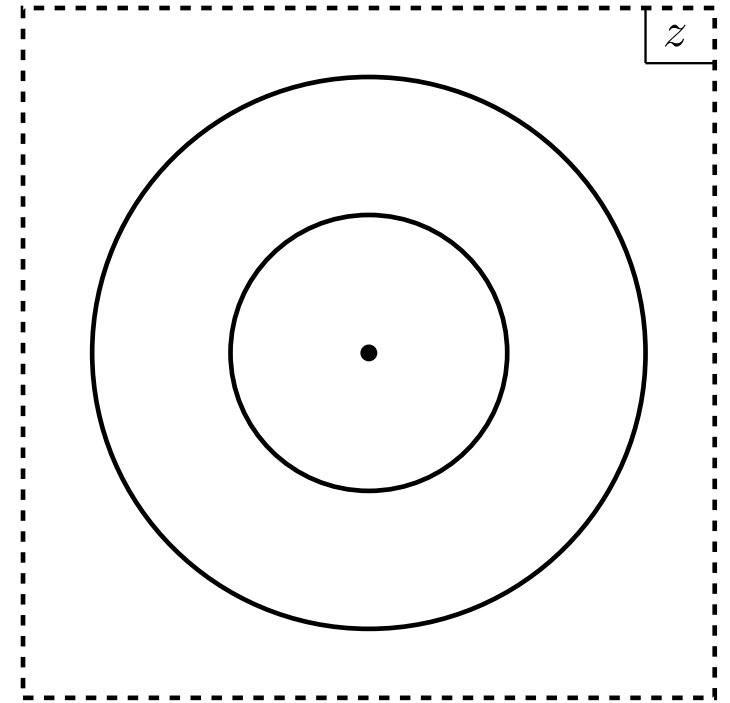
$$p_n |\text{phys.}\rangle = 0$$

holomorphic splitting

Analytically continue and map Lorentzian cylinder to **Riemann sphere**



$$z = e^{\eta_E - i\vartheta}$$

Complex coordinates

with metric

$$ds^2 = \Omega(z, \bar{z})^2 dz d\bar{z}, \quad \Omega(z, \bar{z})^2 = \frac{4}{H^2} \frac{1}{(1 + z\bar{z})^2}$$

Shift transformations generated by

$$\mathcal{F}_{\mu_1 \dots \mu_{k+1}} = \nabla_{(\mu_1} \dots \nabla_{\mu_{k+1})} \phi$$

→ **Shift-invariant** operators

$$F = \nabla_z^{k+1} \phi, \quad \bar{F} = \nabla_{\bar{z}}^{k+1} \phi$$

Conservation = **holomorphicity**

$$\nabla^\lambda \mathcal{F}_{\lambda \mu_2 \dots \mu_{k+1}} = 0 \quad \longrightarrow \quad \bar{\partial} F = 0, \quad \partial \bar{F} = 0$$

Holomorphic factorisation

Map from field to current

$$\nabla^{k+1} \cdot = \left(\partial - 2k \frac{\partial \Omega}{\Omega} \right) \dots \left(\partial - 2 \frac{\partial \Omega}{\Omega} \right) \partial \cdot$$

→ Define “conjugate” operator:

$$\mathcal{D}_k \cdot = \left(\partial + 2 \frac{\partial \Omega}{\Omega} \right) \dots \left(\partial + 2k \frac{\partial \Omega}{\Omega} \right) \cdot$$

which satisfies

$$\nabla^{k+1} \mathcal{D}_k f(z) = \partial^{2k+1} f(z), \quad \overline{\nabla}^{k+1} \mathcal{D}_k f(z) = 0$$

Can show: **General solution** to classical e.o.m.

$$\phi(z, \bar{z}) = \mathcal{D}_k \varphi(z) + \overline{\mathcal{D}}_k \tilde{\varphi}(\bar{z})$$

Mode expansions

Match mode expansions in **canonical quantisation**

$$\varphi(z) \sim \sum_{|n| \leq k} \left(\frac{1}{2} i^{k+n} \frac{x_n}{z^{n-k}} + \frac{(-i)^{k+1+n}}{(k-n)!(k+n)!} \frac{p_n}{z^{n-k}} \log z \right) + i^{k+1} \sum_{|n| > k} \frac{1}{n} \frac{1}{\sqrt{\prod_{j=1}^k (n^2 - j^2)}} \frac{\alpha_n}{z^{n-k}}$$

→ Rescaled oscillator modes

$$\alpha_n = \sqrt{n} a_n, \quad \alpha_{-n} = \sqrt{n} a_n^\dagger, \quad \text{for } n \geq 0$$

Holomorphic current admits:

$$F(z) \sim - \sum_{|n| \leq k} i^{k+1+n} \frac{p_n}{z^{n+k+1}} - i^{k+1} \sum_{|n| > k} \sqrt{\prod_{j=1}^k (n^2 - j^2)} \frac{\alpha_n}{z^{n+k+1}}$$

linear symmetries

Observation: **Linear automorphisms** on the space of oscillator modes

$$Q = \sum_{m,n} A_{mn} a_m^\dagger a_n, \quad \text{arbitrary } A_{mn}$$

are **linear symmetries** of the theory.

dS_2 Isometries

Isometries of background (Riemann sphere) $SO(3) = PSU(2)$: Infinitesimally

$$z \mapsto z + \varepsilon(z), \quad \varepsilon \in \{1, z, z^2\}$$

→ Also: **2d global conformal transformations!**

Idea: Treat currents as primaries with **holomorphic weights**

$$(h_F, \bar{h}_F) = (k + 1, 0) \quad \longrightarrow \quad F'(w) = \left(\frac{\partial w(z)}{\partial z} \right)^{-(k+1)} F(z)$$

Corollary: **Ward identities** fix correlation functions of currents.

$$\langle F(z)F(z') \rangle = (-1)^{k+1} \frac{1}{4\pi} (2k + 1)! \frac{1}{(z - z')^{2k+2}}$$

Oscillator realisation

Do there exist operators s.t. $\delta_{z^{n+1}} \cdot = [\ell_n, \cdot]$?

Yes!

$$\ell_1 = \sum_{n=k+1}^{\infty} \sqrt{\frac{(n-k-1)(n+k)}{n(n-1)}} \alpha_{1-n} \alpha_n + \sum_{n=-(k-1)}^k (k+1-n) x_{1-n} p_n + \frac{\sqrt{2k+1} i^{-k}}{\sqrt{(k+1)(2k)!}} \alpha_{k+1} p_{-k} + \text{anti-hol.}$$

$$\ell_0 = \sum_{n=k+1}^{\infty} \alpha_{-n} \alpha_n + i \sum_{n=-k}^k n x_{-n} p_n + \text{anti-hol.}$$

$$\ell_{-1} = \sum_{n=k+1}^{\infty} \sqrt{\frac{(n-k)(n+k+1)}{n(n+1)}} \alpha_{-1-n} \alpha_n + \sum_{n=-k}^{k-1} (k+1+n) x_{-1-n} p_n + \frac{\sqrt{2k+1} i^k}{\sqrt{(k+1)(2k)!}} \alpha_{-k-1} p_k + \text{anti-hol.}$$

...**non-local** in terms of field

→ Holomorphic + global conformal symmetry: **Local conformal symmetry?**

Local conformal symmetry?

$$\delta_\varepsilon F(z) = -\varepsilon(z) \partial F(z) - (k+1) \partial \varepsilon(z) F(z)$$

Read off action of infinitesimal transformations

Diagram illustrating the action of infinitesimal transformations on the Laurent series of $F(z)$:

$$F(z) \sim \cdots + \alpha_{-k-1} + \frac{p_{-k}}{z} + \frac{p_{-k+1}}{z^2} + \cdots + \frac{p_{k-1}}{z^{2k}} + \frac{p_k}{z^{2k+1}} + \frac{\alpha_{k+1}}{z^{2k+2}} + \cdots$$

Red arrows indicate the action of $\ell_2 \rightarrow \varepsilon \sim z^3$, showing shifts in the coefficients of the series.

Blue arrows indicate the action of $\ell_1 \rightarrow \varepsilon \sim z^2$, showing shifts in the coefficients of the series.

Putative **Virasoro generators** need

$$\ell_m \supset \sum_{n=0}^{m-1} \#x_{k-n} \alpha_{-k-m+n} \longrightarrow [\ell_m, F] \supset \sum_{n=0}^{m-1} \#x_{k-n}$$

→ These appear only in $\varphi(z)$: Transformations necessarily **non-local**

Oscillator realisation

Instead:

$$L_m = \sum_{n=k+m+1}^{\infty} \sqrt{\frac{(n-m-k)(n-k)}{(n-m)n}} \alpha_{m-n} \alpha_n + \frac{1}{2} \sum_{n=1}^{m-1} \sqrt{\frac{(m-n)n}{(k+m-n)(k+n)}} \alpha_{k+m-n} \alpha_{k+n} + \text{anti-hol.}$$

satisfy **Virasoro algebra**

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0}$$

→ Read off central charge from e.g.

$$\langle \Omega | [L_2, L_{-2}] | \Omega \rangle = \frac{1}{2} \quad \longrightarrow \quad c = 1$$

conclusion

Summary

Resemblance of massive fields in dS_2 with massless fields in flat space.

- Existence of (anti-)holomorphic currents
- Mode expansions in complex coordinates
- Holomorphic factorisation
- Non-local linear symmetries
- Hidden conformal symmetry

In the paper:

- AdS_2
- Chiral algebra/Integrable deformations

Coming up...

- Holomorphic scalar as worldsheet theory
- Higher-genus surfaces?
- Fermions: Supersymmetry? Bosonisation?

Thank you for your attention!

Bonus slides

Mode solutions

Fourier decompose w.r.t. angular coordinate

$$f_n(\eta, \vartheta) = \psi_{|n|}(\eta) e^{in\vartheta}, \quad |n| \in \mathbb{Z}$$

$$\left(-\partial_\eta^2 + \frac{k(k+1)}{\cos(\eta)^2} \right) \psi_n(\eta) = n^2 \psi_n(\eta)$$

Physical solutions (not real)

$$\psi_{|n|} \sim e^{-i|n|\eta} P_k^{(-|n|, |n|)}(-i \tan(\eta)), \quad |n| \geq k+1$$

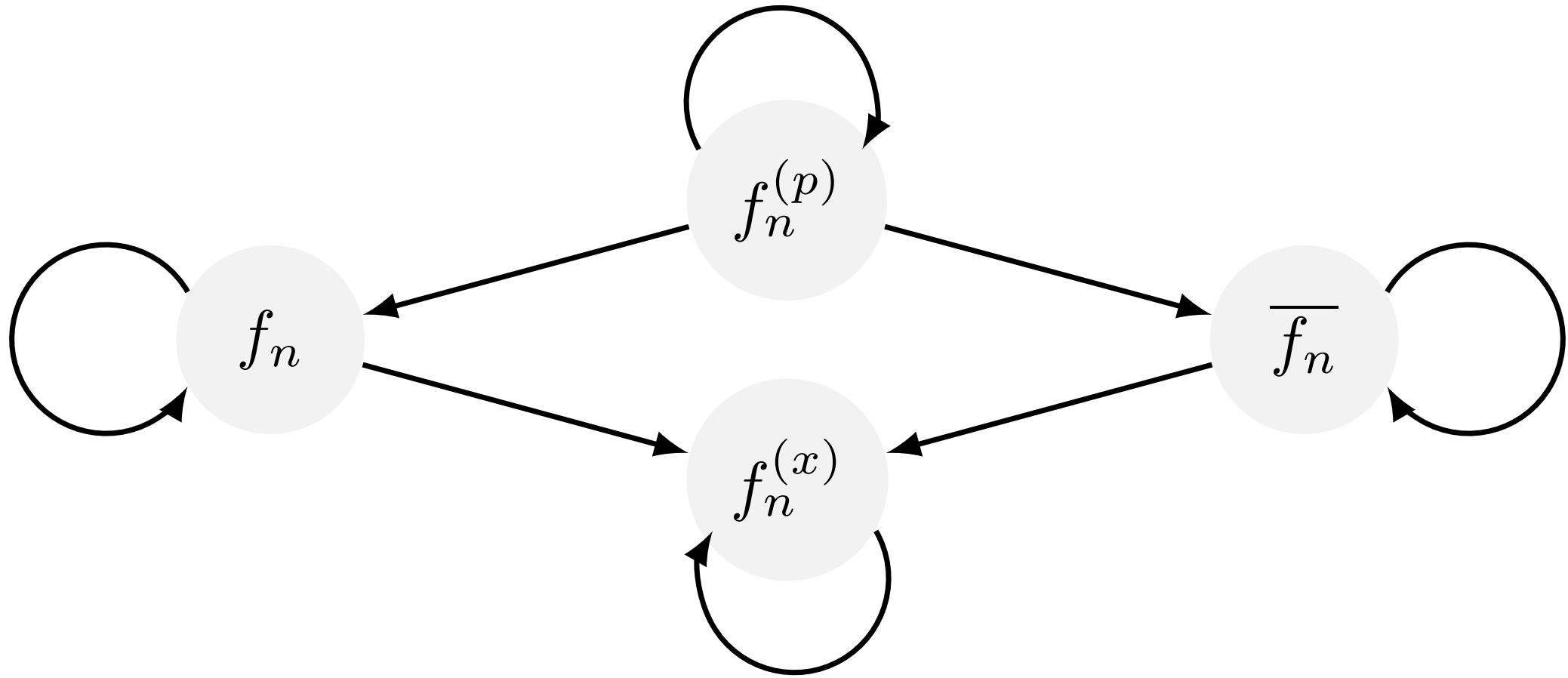
Up to constant, position-modes spanned by taking $|n| \leq k$

Second independent solution

$$g_n \sim \left(\frac{1}{2} \cos(\eta) \right)^{k+1} {}_2F_1 \left(k+1+n, k+1-n; k+\frac{3}{2}; \frac{1-\sin(\eta)}{2} \right) e^{in\vartheta}$$

	x -modes	p -modes
$k = 0$	$\frac{1}{2\sqrt{\pi}}$	$\frac{1}{\sqrt{\pi}}\eta$
$k = 1$	$\frac{1}{2\sqrt{\pi}} \tan(\eta)$ $\frac{1}{2\sqrt{\pi}} e^{\pm i\vartheta} \sec(\eta)$	$-\frac{1}{\sqrt{\pi}} (1 + \eta \tan(\eta))$ $\frac{1}{2\sqrt{\pi}} e^{\pm i\vartheta} (\sin(\eta) + \eta \sec(\eta))$
$k = 2$	$\frac{1}{2\sqrt{\pi}} (1 + 3 \tan(\eta)^2)$ $\frac{3}{2\sqrt{\pi}} e^{\pm i\vartheta} \sec(\eta) \tan(\eta)$ $\frac{3}{2\sqrt{\pi}} e^{\pm 2i\vartheta} \sec(\eta)^2$	$\frac{1}{4\sqrt{\pi}} (\eta + \tan(\eta) + 3\eta \tan(\eta)^2)$ $-\frac{1}{6\sqrt{\pi}} e^{\pm i\vartheta} \sec(\eta) (3 - \cos(\eta)^2 + 3\eta \tan(\eta))$ $\frac{1}{96\sqrt{\pi}} e^{\pm 2i\vartheta} \sec(\eta)^2 (\sin(4\eta) + 8 \sin(2\eta) + 12\eta)$

dS isometries map modes as follows



Differential operators on Riemann surfaces

Complex structure + metric: Tensors classified in terms of **holomorphic rank**

$$h = \text{lower } z \text{ indices} + \text{upper } \bar{z} \text{ indices} - \text{upper } z \text{ indices} + \text{lower } \bar{z} \text{ indices}$$

Operator is not $\mathcal{D}_k$ conjugate w.r.t invariant inner product

$$\langle V, W \rangle = \int dz d\bar{z} \sqrt{|g|} (g^{z\bar{z}})^h V \bar{W}$$

Instead satisfies:

$$\int dz d\bar{z} \sqrt{|g|} \nabla^\alpha f(z, \bar{z}) g(z, \bar{z}) = (-1)^\alpha \int dz d\bar{z} \sqrt{|g|} f(z, \bar{z}) \mathcal{D}_\alpha g(z, \bar{z})$$

In terms of “real” conjugate operator

$$\overline{(\nabla^\alpha)^\dagger} g_{z\bar{z}}^\alpha \equiv (-1)^\alpha g_{z\bar{z}}^\alpha \mathcal{D}_\alpha.$$